

Fault Code: 1418

Gain Adjust Potentiometer Circuit

 Printable Version

Overview

Codes	Reason	Effect
Fault Code: 1418 PID(P): SPN: FMI: Lamp: Warning SRT:	Gain adjust potentiometer signal is shorted high.	The gain adjustment feature will be disabled, and a default value of zero will be used. Possible loss of performance.

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Gain Adjust Potentiometer Circuit

Circuit Description

The gain adjust potentiometer is so the operator can adjust the amplification of the difference in the value between the “desired speed” and the “actual speed” of the engine, so that the governor can respond to engine speed fluctuations. The difference between the “desired speed” and the “actual speed” is called the error signal. When the gain is increased, the error signal amplification is increased, more corrective action is taken by the governor, and its response is quicker. When the gain is reduced, the error signal amplification is reduced, less corrective action is taken by the

governor, and its response is slower. If the gain is set too high, the governor can overcorrect the speed error signal, which will lead to engine speed surging or “hunting.” The ECM monitors the voltage and expects to see the voltage vary between 0.5 and 4.5 volts during normal operation. High voltage will trip Fault Code 1418 and can be caused by shorts in the signal wire, an open in the return wire, or a failed potentiometer.

Component Location

Refer to the OEM manual for location.

Shoptalk

Potentiometers are very sensitive to the environment. Clean the potentiometer, and check its resistance first.

Refer to Troubleshooting Fault Code t05-1418 (</qs3/pubsys2/xml/en/procedures/01/01-t05-1418.html>)